

DATE: EXPT NO:01	EARTHQUAKE ANALYSIS SYSTEM USING MACHINE LEARNING
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AIM

To analyze earthquake data using Exploratory Data Analysis (EDA), visualize important earthquake patterns, preprocess the dataset, build a Machine Learning model to predict tsunami occurrence, evaluate the model, and export the processed data to Excel.

PROCEDURE

1. Download the Earthquake dataset.
2. Import the required Python libraries such as Pandas, Matplotlib, Seaborn, and Scikit-learn.
3. Load the earthquake dataset using Pandas.
4. Display the first few records, dataset shape, information, and missing values.
5. Remove unnecessary columns such as title and location columns.
6. Check and handle missing or duplicate values.
7. Analyze important features such as magnitude, depth, latitude, longitude, cdi, mmi, sig, gap, and nst.
8. Visualize the distribution of tsunami occurrences using a count plot.
9. Create a visualization to identify the relationship between magnitude and tsunami occurrence.
10. Separate the input features (X) and target variable (y), where tsunami is the target.
11. Split the dataset into training and testing data.
12. Train a Machine Learning model using the training data.
13. Predict tsunami occurrence using the test data.
14. Evaluate the model using suitable performance metrics such as Accuracy Score and Confusion Matrix.
15. Export the processed earthquake data to an Excel file.
16. Display the analysis results.

CODE:

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
```

```
from sklearn.metrics import accuracy_score, confusion_matrix

# 1. Load dataset
df = pd.read_csv("earthquake_data.csv")

# 2. Display data
print(df.head())

# 3. Basic information
print(df.shape)
print(df.info())

# 4. Check missing values
print(df.isnull().sum())

# 5. Fill missing values
df["cdi"] = df["cdi"].fillna(df["cdi"].median())
df["mmi"] = df["mmi"].fillna(df["mmi"].median())

# 6. Remove unnecessary columns
df = df.drop(["title", "location"], axis=1)

# 7. Visualization - Tsunami
sns.countplot(x="tsunami", data=df)
plt.title("Tsunami Occurrence")
plt.show()

# 8. Visualization - Magnitude vs Tsunami
sns.countplot(x="magnitude", hue="tsunami", data=df)
plt.title("Magnitude vs Tsunami")
plt.xticks(rotation=20)
plt.show()

# 9. Convert categorical data into numbers
df = pd.get_dummies(df, drop_first=True)

# 10. Separate features and target
X = df.drop("tsunami", axis=1)
y = df["tsunami"]

# 11. Split data
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# 12. Train Logistic Regression model
model = LogisticRegression(max_iter=1000)
model.fit(X_train, y_train)
```

```

# 13. Make prediction
y_pred = model.predict(X_test)

# 14. Calculate accuracy
accuracy = accuracy_score(y_test, y_pred)
print("Accuracy:", accuracy)

# 15. Confusion Matrix
cm = confusion_matrix(y_test, y_pred)
sns.heatmap(cm, annot=True, fmt="d")
plt.title("Confusion Matrix")
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.show()

# 16. Export DataFrame to Excel
df.to_excel("earthquake_analysis.xlsx", index=False)

```

OUTPUT:

mmi	alert	tsunami	sig	net	nst	dmin	gap	magType	depth	latitude	longitude	location	continent	country	title	magnitude	date_time	cdi
0	M 7.0 - 18 km SW of Malango, Solomon Islands								7.0	22-11-2022 02:03								8
7	green	1	768	us	117	0.509	17.0	mww	14.000	-9.7963								159.596
	Malango, Solomon Islands											Oceania	Solomon Islands					
1	M 6.9 - 204 km SW of Bengkulu, Indonesia								6.9	18-11-2022 13:37								4
4	green	0	735	us	99	2.229	34.0	mww	25.000	-4.9559								100.738
	Bengkulu, Indonesia											NaN						
2																		
3	green	1	755	us	147	3.125	18.0	mww	579.000	-20.0508								-178.346
	NaN Oceania																	
3	M 7.3 - 205 km ESE of Neiafu, Tonga								7.3	11-11-2022 10:48								5
5	green	1	833	us	149	1.865	21.0	mww	37.000	-19.2918								-172.129
	Neiafu, Tonga											NaN						
4																		
2	green	1	670	us	131	4.998	27.0	mww	624.464	-25.5948								178.278
	NaN																	
		NaN																

(782, 19)

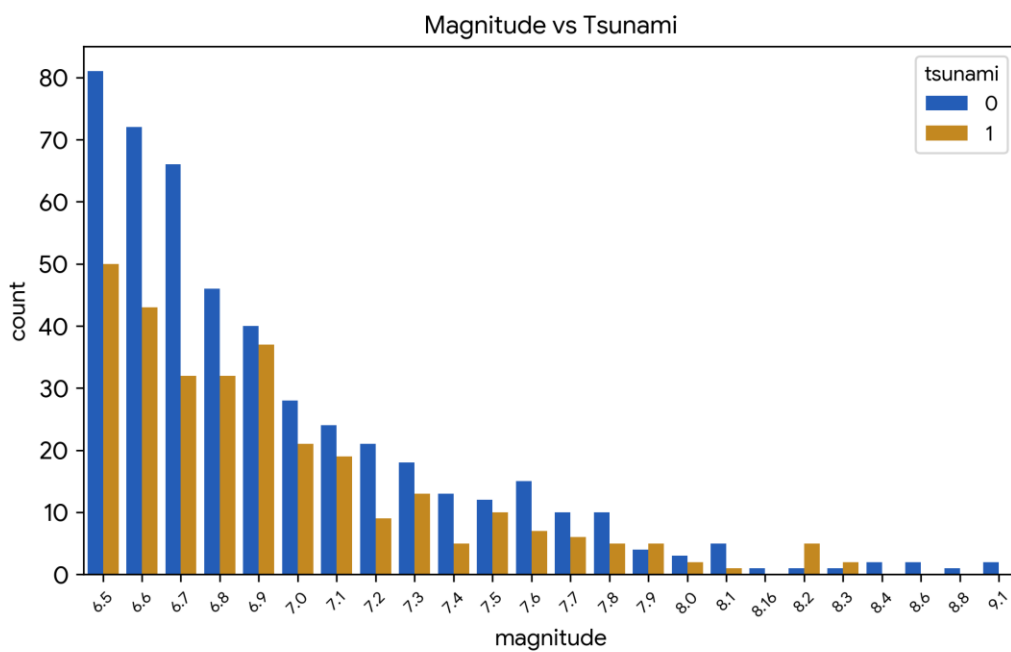
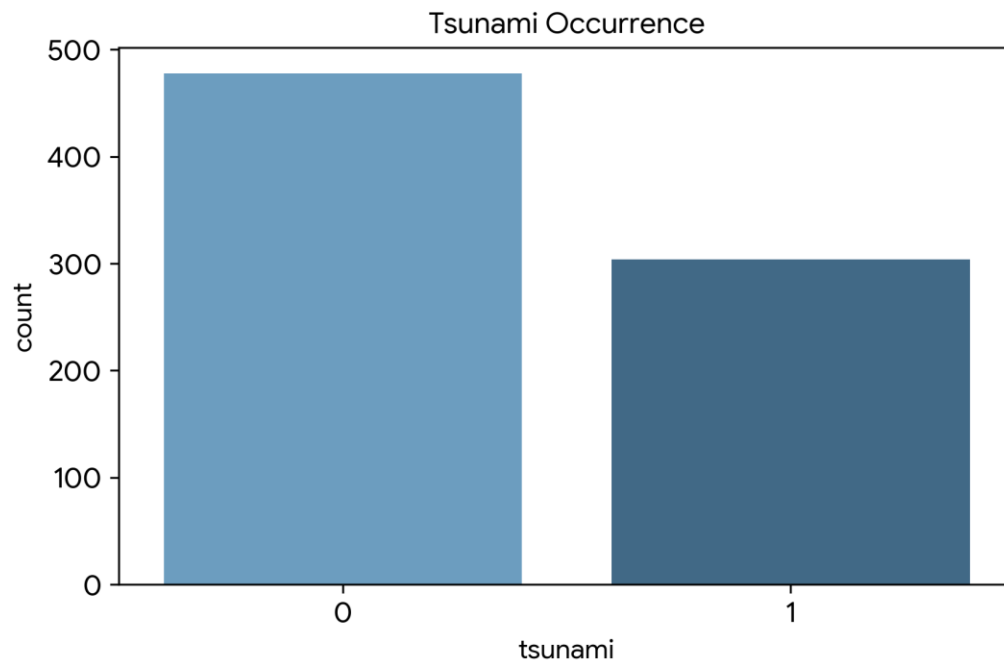
```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 782 entries, 0 to 781
Data columns (total 19 columns):
#   Column      Non-Null Count  Dtype
---  -
0   title       782 non-null   object
1   magnitude   782 non-null   float64

```

```
2  date_time  782 non-null  object
3  cdi        782 non-null  int64
4  mmi        782 non-null  int64
5  alert      415 non-null  object
6  tsunami    782 non-null  int64
7  sig        782 non-null  int64
8  net        782 non-null  object
9  nst        782 non-null  int64
10 dmin       782 non-null  float64
11 gap        782 non-null  float64
12 magType    782 non-null  object
13 depth      782 non-null  float64
14 latitude   782 non-null  float64
15 longitude  782 non-null  float64
16 location   777 non-null  object
17 continent  206 non-null  object
18 country    484 non-null  object
dtypes: float64(6), int64(5), object(8)
memory usage: 116.2+ KB
None
```

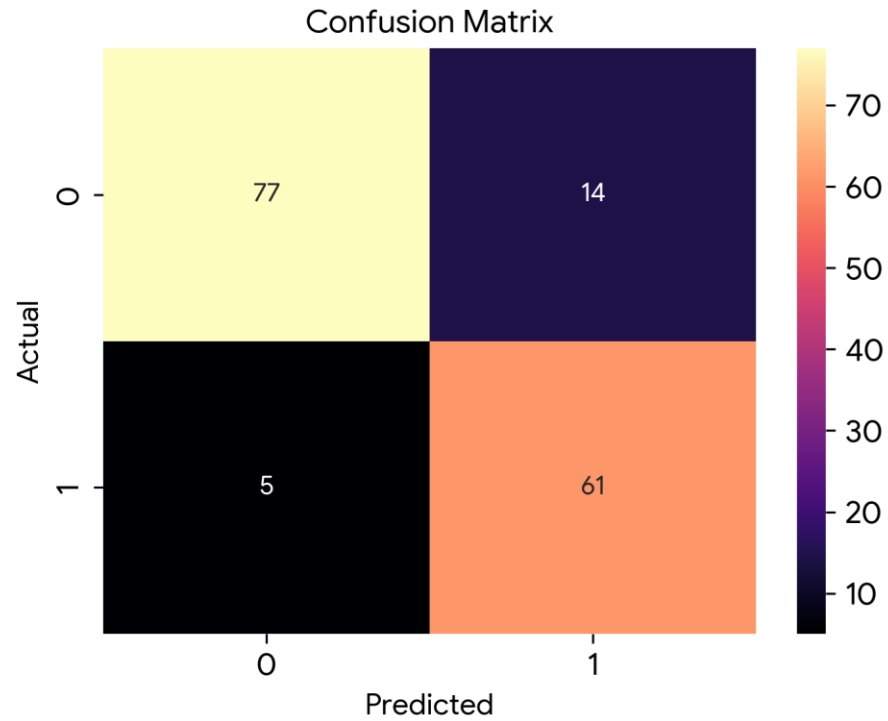
```
title          0
magnitude      0
date_time      0
cdi            0
mmi            0
alert          367
tsunami        0
sig            0
net            0
nst            0
dmin           0
gap            0
magType        0
depth          0
latitude       0
longitude      0
location       5
continent      576
country        298
dtype: int64
```



```

/usr/local/lib/python3.13/dist-packages/sklearn/linear_model/_logistic.py:465:
ConvergenceWarning: lbfgs failed to converge (status=1):
STOP: TOTAL NO. OF ITERATIONS REACHED LIMIT.
Accuracy: 0.8726114649681529

```



RESULT:

The earthquake dataset was successfully imported, analyzed, visualized, and processed. A Logistic Regression model was trained to predict tsunami occurrence, and the model performance was evaluated using accuracy and a confusion matrix. The processed data was also exported to an Excel file.